

UNIVERSITY OF TORONTO MISSISSAUGA  
DECEMBER 2025 FINAL EXAMINATION  
MAT102H5F - Introduction to Proofs  
Masoud Ataei, Ali Feizmohammadi, Tyler Holden, Denis Gorodkov,  
Ivan Khatchatourian  
Duration - 3 hours  
Aids: None

Name: \_\_\_\_\_

Student Number: \_\_\_\_\_

Signature: \_\_\_\_\_

This exam contains 10 pages (including this cover page) and 7 questions. The number of points for each question is indicated in the question itself, with Question 7 being worth 10 marks, making the entire exam out of 42 marks. Check to see if any pages are missing and ensure that all required information at the top of this page has been filled in. You must also have the corresponding Multiple Choice Booklet, which contains the multiple choice questions needed for Question 7.

Each part of Question 7 includes five multiple choice answers, of which only a single response is correct. All answers should be indicated on the included scantron sheet (page 10). When filling out the scantron:

- Use a dark pencil or erasable ink.
- Ensure that bubbles are completely filled and that erased answers are completely erased.

Any failure to follow these instructions which results in the scanner not being able to read your work will result in a deduction of 3 marks. You will not be penalized for incorrect answers.

Your solutions to Questions 1 - 6 must be written in the space provided. There are two additional pages (pages 8 and 9) at the end of the booklet for any overflow work. Clearly indicate in your solution if your work will continue onto that page. No additional paper may be submitted for grading. Do not, under any circumstances, write on the QR code in the top right corner of each page. **Doing so will result in you receiving a zero (0) on this exam.**

The University of Toronto Mississauga and you, as a student, share a commitment to academic integrity. You are reminded that you may be charged with an academic offence for possessing any unauthorized aids during the writing of an exam. Clear, sealable, plastic bags have been provided for all electronic devices with storage, including but not limited to: cell phones, SMART devices, tablets, laptops, calculators, and MP3 players. Please turn off all devices, seal them in the bag provided, and place the bag under your desk for the duration of the examination. You will not be able to touch the bag or its contents until the exam is over.

If, during an exam, any of these items are found on your person or in the area of your desk other than in the clear, sealable, plastic bag, you may be charged with an academic offence. A typical penalty for an academic offence may cause you to fail the course.

Please note, once this exam has begun, you **CANNOT** re-write it.

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1. The following questions are unrelated: Part (i) will not help you with part (ii).
  - (i) (3 points) Suppose that  $f : X \rightarrow Y$  is an injective function. If  $A, B \subseteq X$  and  $f(A) = f(B)$ , show that  $A = B$ .
  - (ii) (3 points) Suppose that  $A$ ,  $B$ , and  $C$  are sets such that  $|A| = |B|$  and  $|B| \leq |C|$ . Prove that  $|A| \leq |C|$ .

2. (5 points) Recall that the Fibonacci sequence is defined as  $F_1 = F_2 = 1$  with  $F_{n+1} = F_n + F_{n-1}$  for all  $n \geq 2$ . Show that for all  $n \geq 10$ ,

$$\left(\frac{3}{2}\right)^n < F_n < 2^n.$$

*Hint:* You may freely use the fact that  $(3/2)^{10} < 87$ , without proof.

3. For a positive natural number  $n$ , define  $n!$  to be  $n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$ , the product of the first  $n$  positive natural numbers.

(i) (3 points) Let  $p$  and  $q$  be distinct prime numbers, and set  $d = pq$ . Find the smallest  $n$  such that  $d \mid n!$ .

(ii) (3 points) Determine the remainder when  $15!$  is divided by 17.

4. Let  $E \subseteq (0, \infty)$  be a nonempty set satisfying the following properties:

(E1)  $1 \notin E$ .

(E2) If  $u, v \in E$ , then both  $uv \in E$  and  $u^v \in E$ .

(E3) For every real number  $t > 0$  with  $t \neq 1$ , exactly one of  $t \in E$  or  $1/t \in E$  holds (but not both).

You may assume that such a set  $E$  exists. Define a relation  $\triangleright$  on  $(0, \infty)$  by  $x \triangleright y$  if and only if  $y/x \in E$ .

(i) (3 points) Prove that  $\triangleright$  is a strict partial order (irreflexive, transitive, and anti-symmetric).

(ii) (3 points) Prove that if  $x \triangleright y$  and  $r > 1$ , then  $x^r \triangleright y^r$ . *Hint:* Break this into the cases where  $r \in E$  and  $1/r \in E$ .

More space for Q4(ii) located on the next page.

Question 4.(ii) Continued.

5. (3 points) Suppose that  $G$  is a group in which  $(ab)^2 = a^2b^2$  for all  $a, b \in G$ . Show that  $G$  is abelian.

6. Let  $G$  and  $H$  be groups and  $\phi : G \rightarrow H$  be a group homomorphism.

(i) (3 points) Prove that  $\ker(\phi) \leq G$ .

(ii) (3 points) A subgroup  $N \leq G$  is said to be *normal* if  $gng^{-1} \in N$  for all  $g \in G$  and  $n \in N$ . Show that  $\ker(\phi)$  is a normal subgroup of  $G$ .

This page is for additional work and will not be marked unless indicated.



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End of Exam Questions.